

RUSH CITY, MN, USA

WMO: 726679

Lat: 45.698N

Lon: 92.953W

Elev: 923

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-9.2	-21.8	1.7	-13.5	-17.5	2.2	-8.4	18.7	18.3	17.0	18.4	3.3	260	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	88.1	72.7	84.5	70.7	81.9	69.0	75.2	84.1	73.2	81.8	71.3	79.3	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	124.0	81.2	70.3	115.4	79.2	68.3	107.7	76.9	39.3	84.6	37.4	81.9	35.7	79.2	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.4	14.2	DB	-21.5	93.2	5.2	3.6	-25.3	95.8	-28.4	97.9	-31.3	99.9	-35.1	102.5	(4)
(5)			WB	-21.9	78.3	5.2	2.0	-25.6	79.7	-28.6	80.9	-31.5	82.1	-35.2	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	14.3	16.9	30.1	43.6	55.2	64.8	69.9	67.2	59.9	46.2	33.1	20.1		
	DBStd	21.69	13.17	11.99	12.57	9.86	8.53	6.18	5.44	5.78	8.62	9.12	10.28	11.30		
	HDD50	4577	1108	926	626	237	42	0	0	0	15	184	511	928		
	HDD65	8237	1573	1346	1083	642	324	76	17	41	200	586	956	1393		
	CDD50	2237	0	0	9	46	202	445	617	533	313	67	5	0		
	CDD65	420	0	0	0	2	19	71	168	109	47	4	0	0		
	CDH74	4036	0	0	3	29	258	705	1602	977	420	42	0	0		
	CDH80	1072	0	0	0	5	70	162	463	258	110	4	0	0		
(14) Wind	WSAvg	5.9	6.0	6.3	6.5	7.4	6.6	5.4	4.6	4.3	5.4	6.2	6.3	5.9		
(15) Precipitation	PrecAvg	31.7	0.9	0.9	1.5	2.8	3.5	4.3	4.1	4.1	3.7	3.1	1.8	1.1		
	PrecMax	39.5	2.2	1.9	2.9	8.3	7.1	7.2	7.2	10.9	6.8	8.0	4.7	2.1		
	PrecMin	23.3	0.1	0.1	0.6	0.6	1.0	2.4	1.5	1.4	0.9	1.1	0.0	0.3		
	PrecStd	4.9	0.6	0.5	0.7	1.6	1.7	1.3	1.5	2.1	1.9	1.8	1.4	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	49.8	69.0	78.7	87.9	89.7	92.0	90.8	88.4	79.5	64.4	45.9		
		MCWB	37.9	44.1	57.2	61.5	68.1	74.1	76.0	73.9	71.5	63.3	54.2	41.6		
	2%	DB	37.5	41.2	59.8	71.3	80.8	84.2	88.1	86.2	82.4	72.6	58.5	40.8		
		MCWB	34.0	36.2	51.5	54.8	63.9	69.2	74.1	72.2	69.5	59.7	50.2	37.5		
	5%	DB	34.6	36.8	53.1	65.9	75.5	81.2	84.8	82.2	79.2	66.0	53.0	36.9		
		MCWB	32.5	33.1	46.3	51.4	60.6	67.7	71.8	69.4	67.1	56.2	46.1	34.4		
	10%	DB	32.1	33.6	46.3	61.3	71.4	78.2	82.0	79.5	74.8	61.2	48.0	34.4		
		MCWB	30.4	30.9	40.7	48.6	58.9	65.4	69.9	67.8	64.8	53.2	43.2	32.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	44.1	60.1	62.5	72.2	76.3	79.1	76.3	75.0	67.5	56.3	43.8		
		MCDB	43.0	50.9	66.1	76.5	82.7	85.5	88.2	86.8	82.8	73.0	61.9	45.5		
	2%	WB	34.9	37.1	52.4	56.8	66.4	72.7	75.9	74.1	71.5	61.3	51.6	37.3		
		MCDB	37.4	40.7	58.4	67.0	75.9	81.5	84.8	82.8	79.8	69.6	57.5	40.0		
	5%	WB	32.8	33.4	46.3	53.0	63.7	69.6	73.9	71.8	68.9	57.5	46.4	34.9		
		MCDB	34.4	36.3	52.9	63.6	72.2	77.8	82.4	79.6	76.5	65.7	51.2	36.9		
	10%	WB	30.2	31.1	41.4	49.9	60.4	67.3	71.8	69.6	66.5	53.8	43.2	32.8		
		MCDB	31.9	33.0	46.2	60.2	68.7	75.5	80.0	77.2	73.7	60.2	48.1	34.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	18.0	18.2	20.5	22.0	20.9	21.1	21.1	21.2	18.1	15.5	13.7		
		MCDBR	16.2	19.0	23.9	29.5	28.4	24.1	23.7	24.3	24.8	26.3	23.2	15.1		
	5% WB	MCWBR	13.6	14.7	15.5	16.9	14.5	12.5	11.9	12.1	13.0	15.2	15.5	12.4		
		MCWBR	15.3	17.6	22.7	25.8	23.6	20.9	21.2	21.5	21.4	22.6	21.1	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.269	0.316	0.371	0.404	0.389	0.384	0.396	0.368	0.316	0.287	0.263		
		taud	2.398	2.361	2.358	2.304	2.258	2.335	2.375	2.352	2.405	2.539	2.513	2.421		
	Ebn at Noon		274	292	291	281	274	279	279	271	269	272	262	259		
		Edh at Noon	22	28	33	39	42	39	37	36	32	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		533	880	1230	1472	1692	1868	1977	1648	1270	791	499	371		
	RadStd		59	81	108	138	136	136	68	123	88	105	45	46		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page